Vishal Vaibhav.

Applied Data Scientist · Retail pricing, promotion & customer analytics

vishalvaibhav121099@gmail.com
vishalvaibhav1210.github.io/portsume
linkedin.com/in/vishal-vaibhav-634898155
Gurugram, India

PROFILE

Applied data scientist on dunnhumby's Technical Solutions team, owning the price, promotion and assortment solutions that retailers use to make weekly commercial decisions. I take models from commercial question to multi-market production: demand and promotion modelling, constrained optimisation, Spark pipelines with data-quality guardrails, and LLM summaries that turn model output into decisions.

EXPERIENCE

Applied Data Scientist · dunnhumby

Nov 2023 – present · Gurugram

Technical owner for price optimisation, promotion analytics and assortment planning: enhancements, production deployment and multi-market scaling.

- Rebuilt a promotion-optimisation system (baseline, promo effect, stockpiling, cannibalisation) as modules with distributed execution, running from weekly promo recommendations up to portfolio investment optimisation; **cut delivery SLA by 70%.**
- Added incremental unit-impact logging and runtime improvements to price and investment optimisation, so teams screen scenarios before full runs finish.
- Built a multi-market demand-parameter precompute pipeline (automated ingestion, Spark, modular ETL orchestration), removing a dependency on external execution teams.
- Embedded data validation (time-window completeness, schema consistency, quality guardrails) into model input pipelines, reducing downstream QA rework.
- Engineered a product recommendation engine combining substitution and adjacency logic, LDA topic models, customer segments and seasonality for agentic commerce workflows.
- Developed training pipelines (Lasso, logistic regression, LDA) for promotion analytics and segmentation, with LLM summaries that turn model output into executive decision briefs.

Manager, LMS & Content · Securitas Group

Jan – Nov 2023 · Gurugram

- Designed and built a Teams chatbot (Node.js, Bot Framework) on the LearnUpon LMS API: a rolling 30-day training calendar where staff register, enrol and unenrol in courses and certifications from chat.
- Built the caching layer that kept the bot responsive under LMS API rate limits; analysed participation data and built reporting on programme effectiveness.

INDEPENDENT SIMULATION STUDIES · synthetic grocery retailer, scored against ground truth, not client work · code on GitHub

- **Price optimisation.** Empirical-Bayes shrinkage of per-product elasticities plus SLSQP under ±10% guardrails captures 92% of the oracle's margin gain (raw estimates: 74%); wrong-sign elasticities fall from 9% to 0.
- **Promotion uplift.** Decomposed 1,488 promotions into gross lift, cannibalisation and pull-forward; only 10%-off deals add margin on average, with supplier-funding sensitivity reported.
- **Range review.** Demand transfer estimated from out-of-stock weeks loses 19% less revenue than delisting the lowest sellers.
- **Customer lifetime value.** BG/NBD and Gamma-Gamma fitted by MLE from scratch; the model's top decile captures 62% of next half-year spend (run-rate: 49%).
- **Need states.** LDA on baskets recovers each shopper's mission mix (r = 0.90) and finds 54% have a real second mission.

SKILLS

Retail science	Elasticity & price optimisation, promotion uplift, forecasting, CLV, segmentation, assortment, recommendations
Methods	Regularised regression, Bayesian shrinkage, BG/NBD, LDA, constrained optimisation, test design
Data & eng.	Python, SQL, PySpark, pandas, NumPy, scikit-learn, SciPy, ETL orchestration, data validation, Node.js
GenAI	LLM summaries of model output, agentic commerce workflows, conversational bots

EDUCATION

B.E., Birla Institute of Technology, Mesra
2017 – 2021 · First Class with Distinction

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification · DeepLearning.AI, 2025
Linear Regression with Python · Coursera, 2025
Python Project for Data Science · Coursera, 2025